Assignment 2

Expectations and Random Vectors

Part I: Theoretical Problems

1. Randomized Depth-First Search

Let X_1 represent the depth of a search tree in a randomized Depth-First Search algorithm, and X_2 represent the number of backtracking steps. Their joint PMF is given by

$$f(x_1, x_2) = \frac{e^{-\lambda}\lambda^{x_1}}{x_1!}\binom{x_1}{x_2}p^{x_2}(1-p)^{x_1-x_2}, \qquad 0 \leq x_2 \leq x_1.$$

(a) Derive the marginal PMFs of X_1 and X_2.

(b) Define $Y = X_1 - X_2$. Prove that X_2 and Y are independent Poisson random variables.

2. Catching the Greedy Counterfeiter

The king's minter boxes his coins n to a box. Each box contains m false coins. The king suspects the minter and randomly draws 1 coin from each of n boxes and has these tested. What is the chance that the sample of n coins contains exactly r false ones? What distribution does it look like, as n approaches ∞?

3. Network Server Load

Let X_1 and X_2 be continuous random variables representing the processing time and network latency of a task, respectively. Their joint PDF is

$$f(x_1, x_2) = \begin{cases} ce^{-x_1}, & 0 \leq x_2 \leq x_1 < \infty, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Determine the normalizing constant c.

(b) Find the conditional PDF $f_{X_2|X_1}(x_2|x_1)$.

(c) Calculate $\mathbb{P}(X_2 > 1 \mid X_1 = 2)$.

4. Total Variance in Trading

The daily number of trades executed by a trading algorithm is $N \sim \text{Poisson}(\lambda)$. The profit from the ith trade is X_i, where the X_i are i.i.d. with mean μ and variance σ^2.

Assume N and all X_i are mutually independent.

(a) Let

$$S = \sum_{i=1}^{N} X_i,$$

with $S = 0$ when $N = 0$. Derive exact closed-form expressions for $\mathbb{E}[S]$ and $\mathrm{Var}(S)$.

5. Uncorrelatedness and Independence

Let X_1 and X_2 be random variables with finite variance.

(a) If $\mathrm{Var}(X_1) = \mathrm{Var}(X_2)$, prove that

$$U = X_1 + X_2, \qquad V = X_1 - X_2$$

are uncorrelated.

(b) Does uncorrelatedness imply independence? Give a rigorous proof or a counterexample.

6. Molina's Urns

Two urns contain the same total numbers of balls, some blacks and some whites in each. From each urn are drawn n (≥ 3) balls with replacement. Find the number of drawings and the composition of the two urns so that the probability that all white balls are drawn from the first urn is equal to the probability that the drawing from the second is either all whites or all blacks. Is it even possible?

7.(optional) Continuous Support over a Unit Disk

Let (X_1, X_2) be a two-dimensional continuous random vector uniformly distributed over the unit disk $D = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 \leq 1\}$.

(a) Formulate the joint PDF $f(x_1, x_2)$.

(b) Derive the marginal PDF $f_{X_1}(x_1)$.

(c) Calculate the conditional expectation $\mathbb{E}[X_2 \mid X_1 = x_1]$.

(d) Prove that the covariance $\mathrm{Cov}(X_1, X_2) = 0$, demonstrating that the variables are uncorrelated.

(e) Using the definition of marginal supports, prove mathematically that X_1 and X_2 are not independent.

8.(optional) Expected Loss in Linear Regression

Let (X_1, X_2, Y) be a three-dimensional continuous random vector, where X_1 and X_2 are predictive features and Y is a target variable. Assume all variables have been centered such that their expected values are zero. Let the covariance matrix elements be denoted by $\mathrm{Cov}(X_i, X_j) = \Sigma_{ij}$, $\mathrm{Cov}(X_i, Y) = C_i$, and let $\mathrm{Var}(Y) = \sigma_Y^2$.

A machine learning model makes the linear prediction $\hat{Y} = w_1 X_1 + w_2 X_2$. The expected Mean Squared Error (MSE) of this model is defined as the expected value of the quadratic loss:

$$J(w_1, w_2) = \mathbb{E}\left[(Y - \hat{Y})^2\right].$$

(a) Expand the expected loss $J(w_1, w_2)$ entirely in terms of the weights w_1, w_2, the covariance components $\Sigma_{11}, \Sigma_{22}, \Sigma_{12}, C_1, C_2$, and the target variance σ_Y^2.

(b) By evaluating the partial derivatives $\frac{\partial J}{\partial w_1}$ and $\frac{\partial J}{\partial w_2}$ and setting them to zero, derive the exact system of linear equations that must be solved to find the optimal predictive weights w_1^* and w_2^*.

Part II: Computational and Simulation Problems

Simulation 1: Empirical Validation of Total Variance

1. Simulate $S = \sum_{i=1}^{N} X_i$ with $N \sim \text{Poisson}(50)$ and $X_i \sim \mathcal{N}(5, 4)$.
2. Generate 100,000 instances and calculate empirical mean and variance.
3. Compare results against the closed-form theory derived in Problem 4.